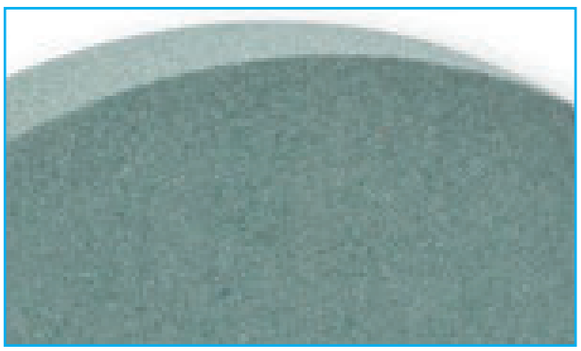
Artificial Abrasives

Artificial abrasives are classified as follows

1. Silicon carbide abrasives
2. Aluminium oxide abrasives.

Silicon carbide abrasives

Silicon carbide manufactured from 56% of silica, 34% of powdered coke, 2% of salt and 12% of saw dust in an electric furnace that is built of loose brick work. There are two types of silicon carbide abrasives green grit and black grit.



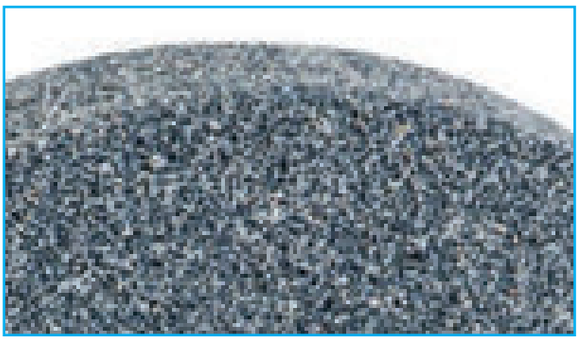
Silicon carbide

Silicon carbide is next to diamond in order of hardness, but it is not tough enough aluminium oxide. It is used for grinding materials of low tensile strength such as cemented carbides, ceramic materials, grey brass, bronze, copper, aluminium, vulcanized rubber etc. This is manufactured under trade names of carborundum. It is denoted by the letter 'S'.

Aluminum oxide

Aluminium oxide is manufactured by heating mineral bauxite, silica, iron oxide, titanium oxide etc mixed with ground coke and iron borings in an electric furnace.

Aluminum oxide is tough and not factured, so it is better adopted to grinding materials of high tensile such as most steels, carbon steels, high speed steels and tough bronzes. This is denoted by letter 'A'.



Aluminum oxide

Types of bond

A bond is an adhesive substance that is employed to hold abrasive grain together in the form of grinding wheels. There are several types of bonds. Different grinding wheels are manufactured by mixing hard abrasives with suitable bonds. The table containing the types of wheel manufactured using different types of bond and their symbols in given below.

Type of bond	Symbol	Grinding Wheel
Vitrified	V	Vitrified wheel
Silicate	S	Silicate wheel
Shellac	E	Elastic wheel
Resinoid	B	Resinoid wheel
Rubber	R	Vulcanized wheel
Oxychloride	O	Oxychloride wheel

Grit, Grade and structure

The grinding wheel is made of thousands of abrasives grains. The grain size or grit number indicated the size of the abrasive grains used in making a wheel or the size of the cutting teeth.

Grain size is denoted by a number indicating the number of meshes per

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